Session-02
Power BI Security and Teams

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Who am I

- Robert Pilgrim, Ph.D. | rpilgrim@uark.edu
- Associate Director of Data Strategy & Insights
- Office of Sponsored Programs, UARK
- B.Sc Astrophysics | Ph.D. Space Science

Connect

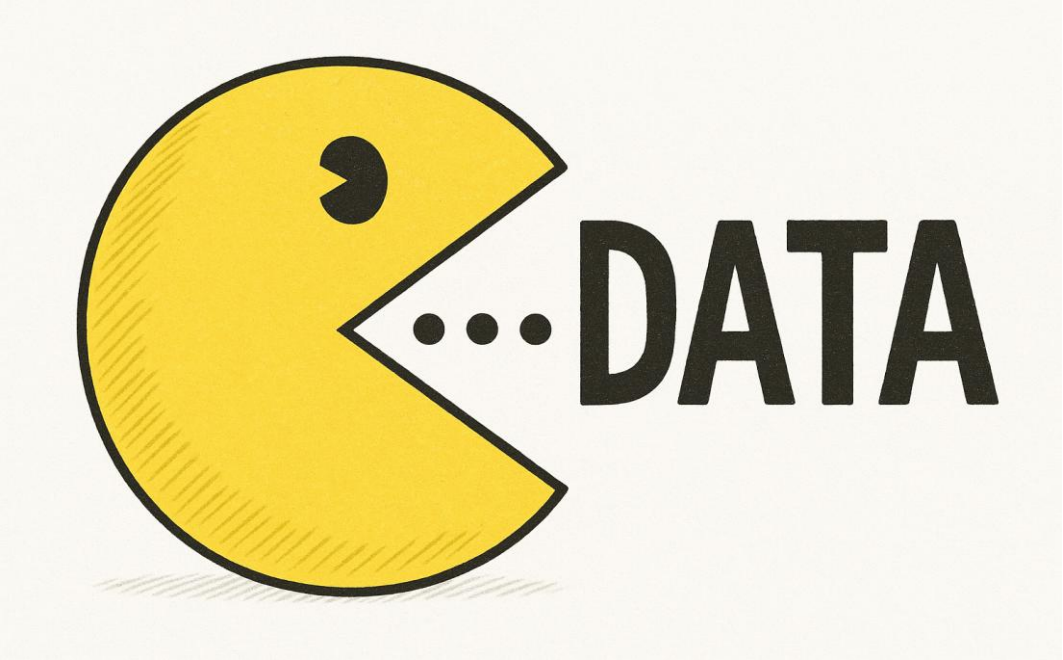
rpilgrim.com

LinkedIn

Substack

Purpose of these Webinars

- Met many people:
  - Conferences / Summits
  - Workshops / presentations
  - REACH mentoring
- Most people just need help starting



Session Framework

One: Power BI Overview

- Introduction
- Microsoft Analytics Core Concepts
- Start on our Power BI Report

Two: Security and Teams

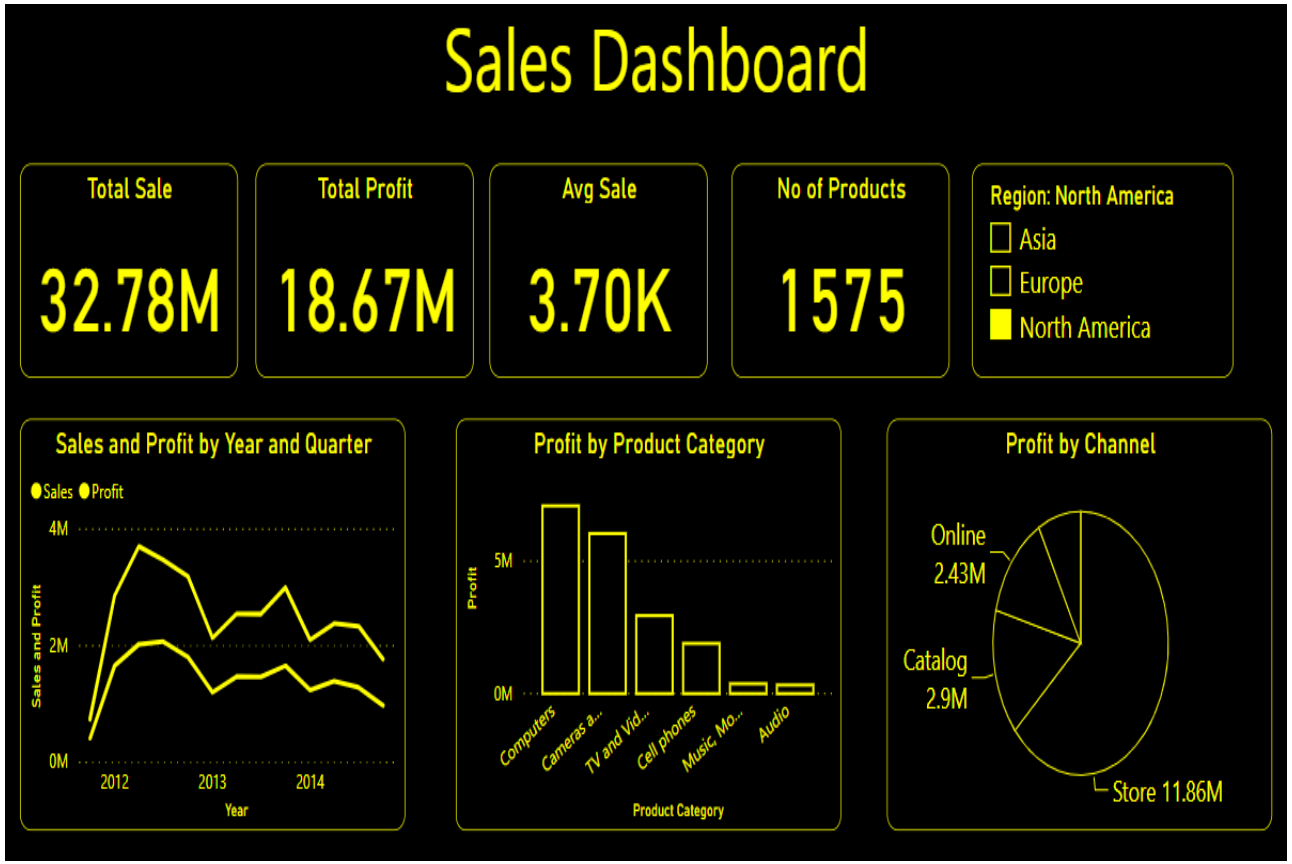
- Continue report
- Deploy Report
- Sharing & Security
- Version Control
- Documentation

Three: Advanced Concepts

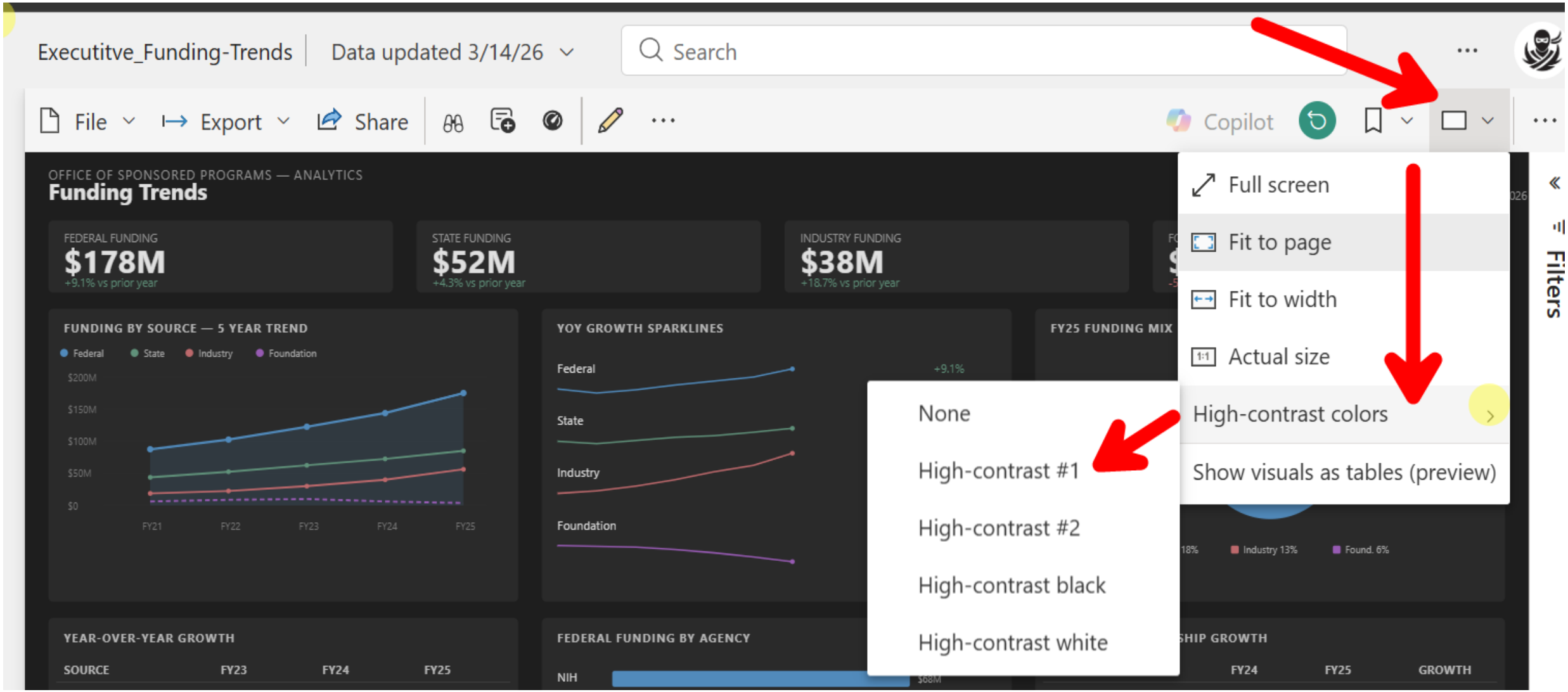
- Finish our Report
- User Defined Functions & Parameters
- R and Python
- External Tools

Question from the Chat — Accessibility

- Accessibility Features
- [Design Power BI Reports for Accessibility (Microsoft Learn)]
- No configuration needed
- Keyboard navigation · Screen-reader compatibility
- High contrast colors, Focus mode
- Show visuals as data table

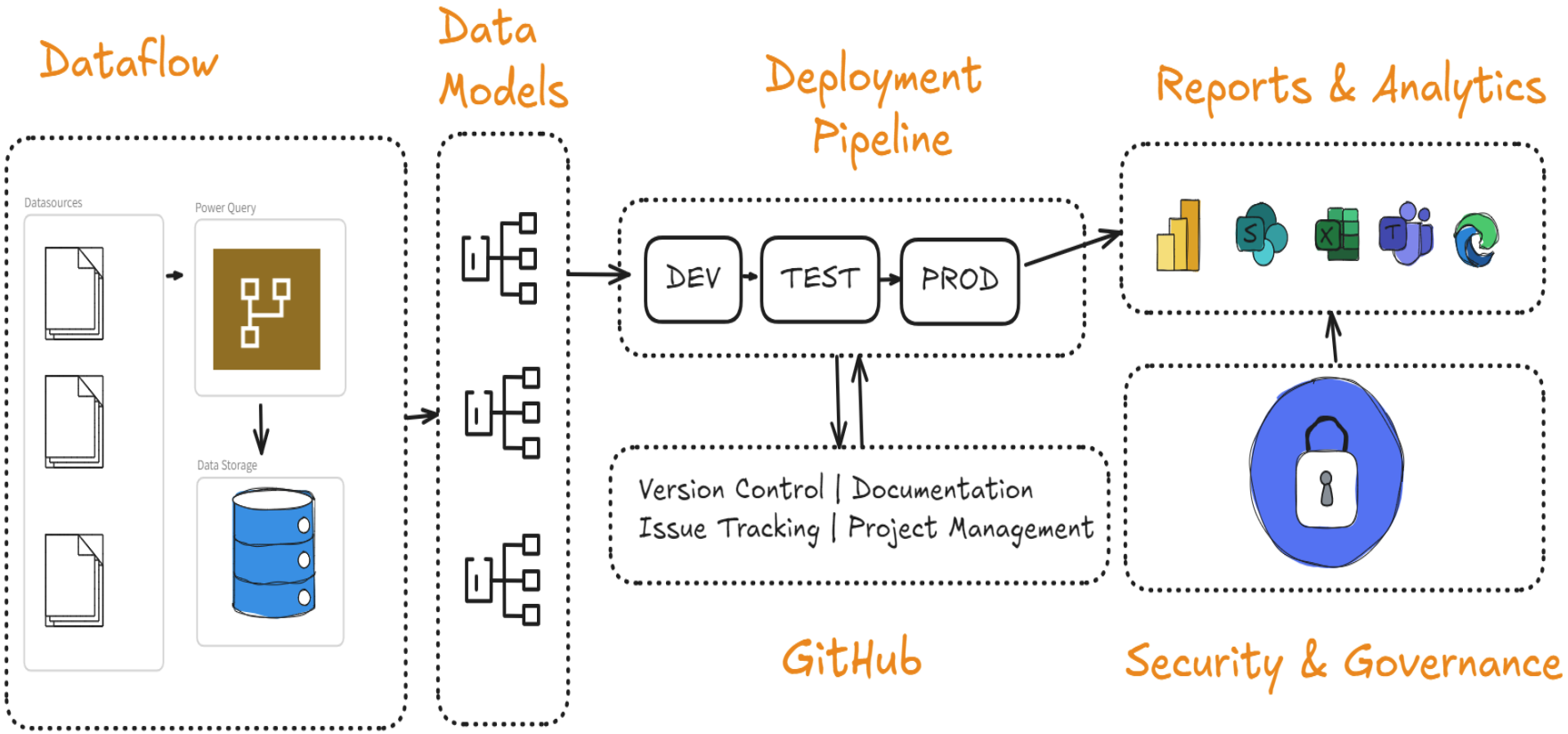


Accessibility



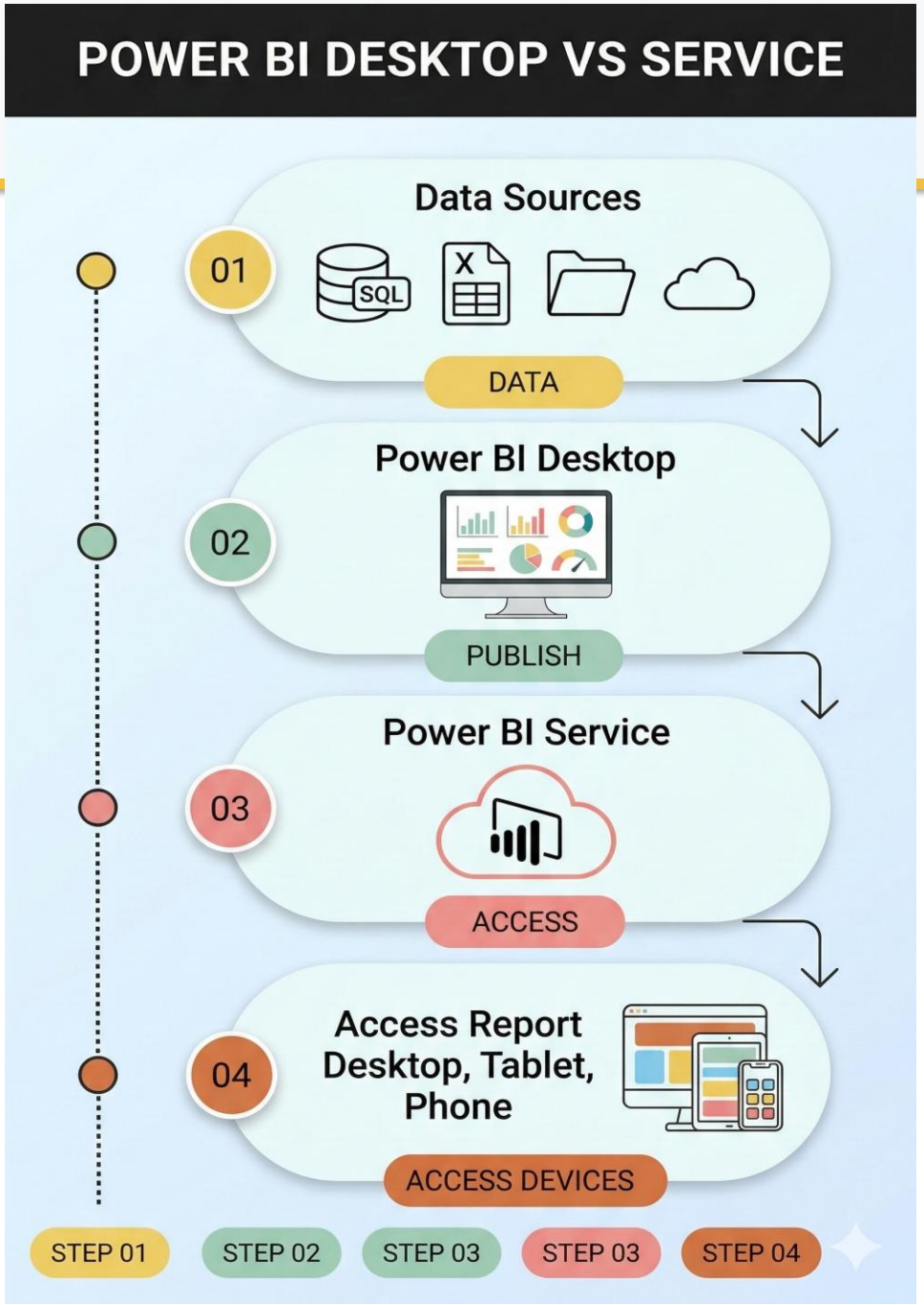
Where We Left Off

- Power BI Desktop, Service, Data Models
- Data Model | DAX



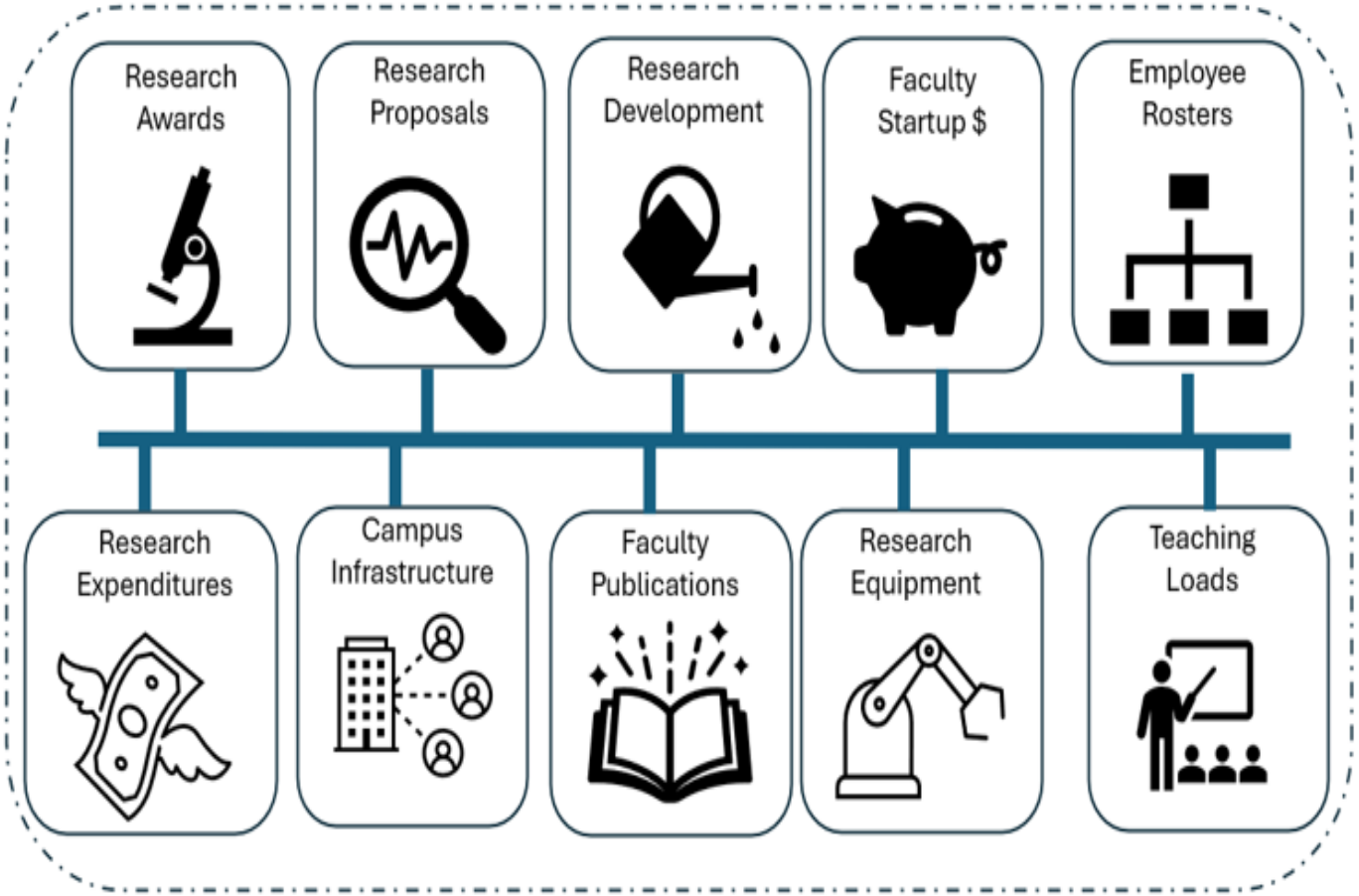
Recap Power BI Service

- Cloud based
- Share / Collaborate
- Microsoft campus?
- app.powerbi.com



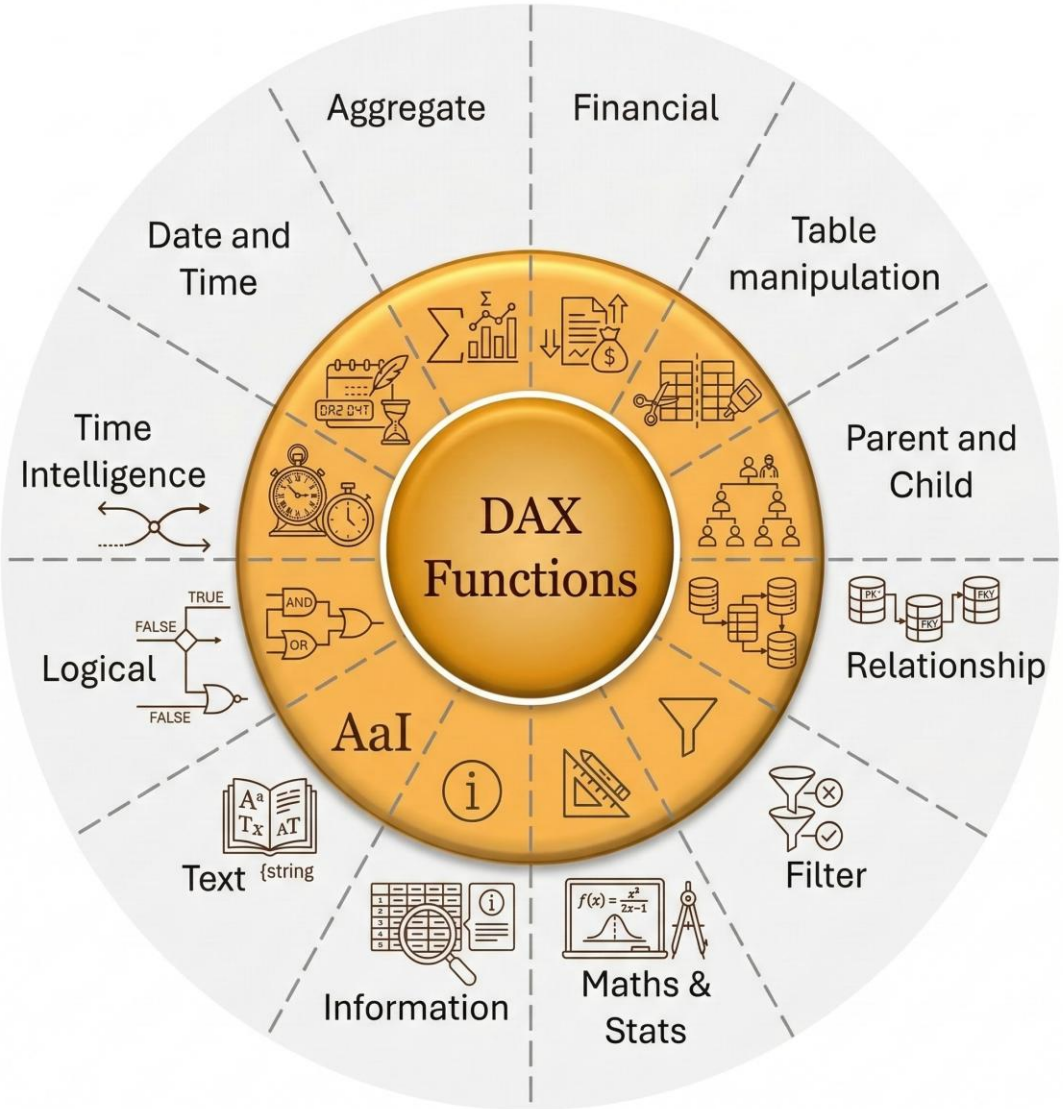
Recap The Data Model

- Relationships NOT merging
- Powered by Data Analysis Expressions (DAX)
- DAX wraps around your data model
- Cross-data analytics



Recap Data Analysis Expressions (DAX)

- Data Models = BRAINS | DAX = LOGIC
- DAX wraps around the data model



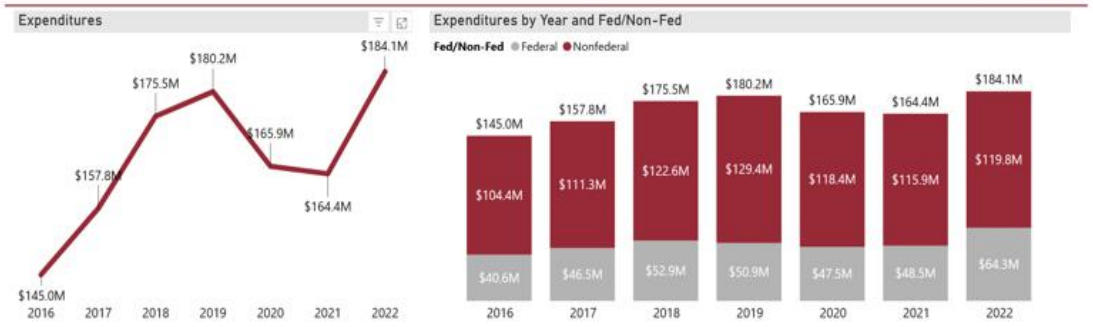
Data Models and Reporting

Data Model

- Design
- Create

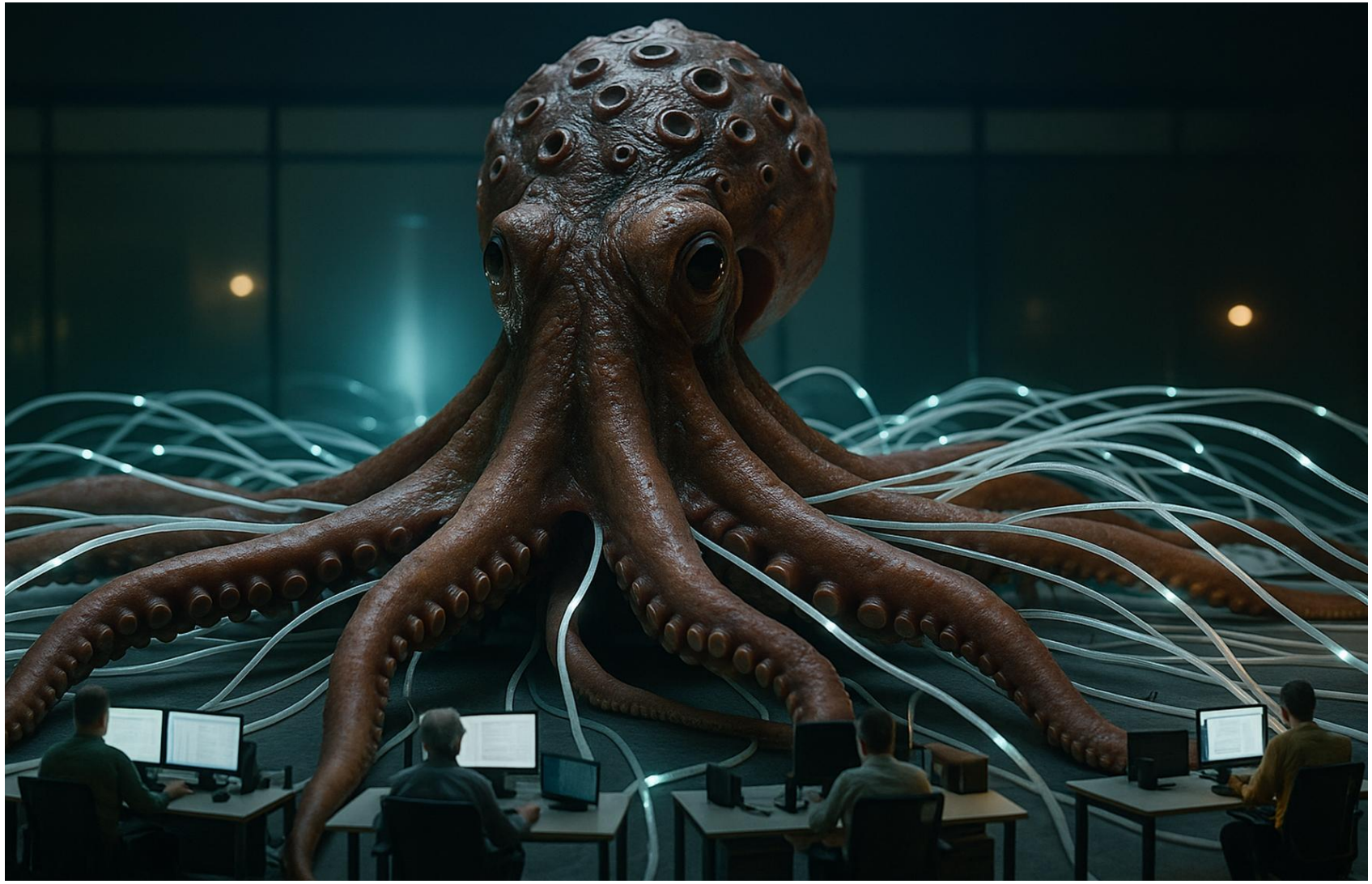
Reporting

- Create
- Distribute



Fed/Non-Fed	2016	2017	2018	2019	2020	2021	2022	Total
Federal	$40,642,000	$46,490,000	$52,937,000	$50,875,000	$47,454,000	$48,534,000	$64,252,000	**$351,184,000**
Nonfederal	$104,378,000	$111,301,000	$122,561,000	$129,350,000	$118,433,000	$115,909,000	$119,828,000	**$821,760,000**
Total	**$145,020,000**	**$157,791,000**	**$175,498,000**	**$180,225,000**	**$165,887,000**	**$164,443,000**	**$184,080,000**	**$1,172,944,000**

Single Source of Truth

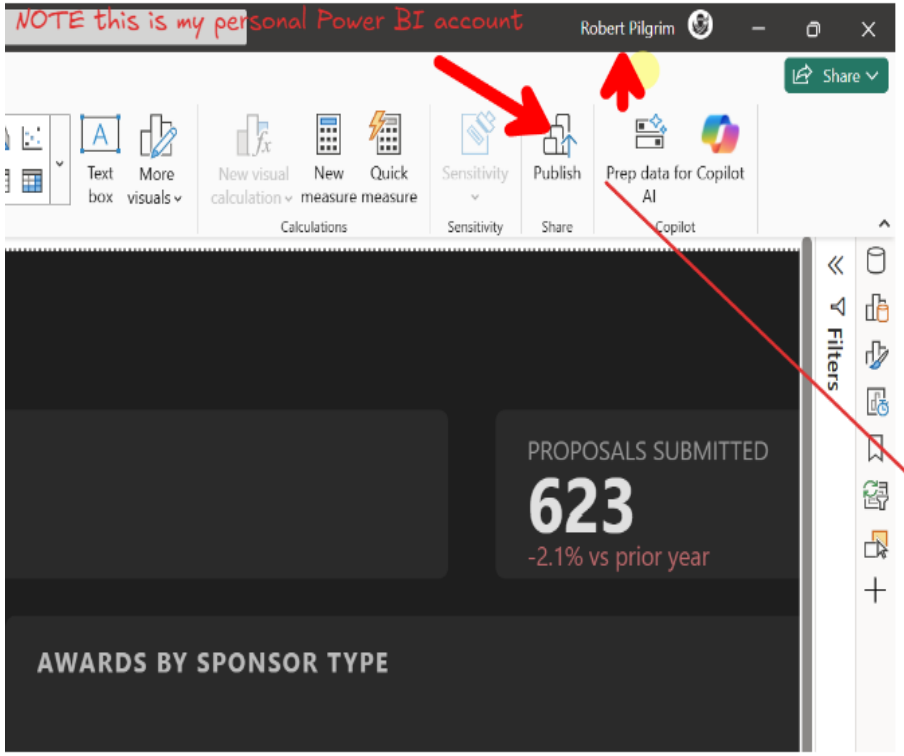


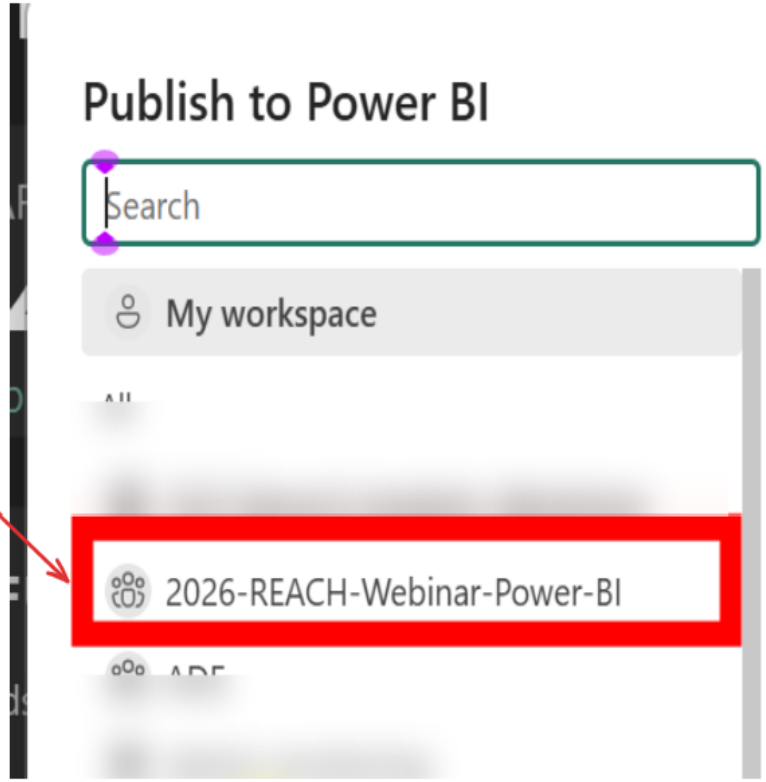
Workspaces

- What is a workspace?
- Creating and organizing workspaces
- Dev / Prod separation

Publishing a Report

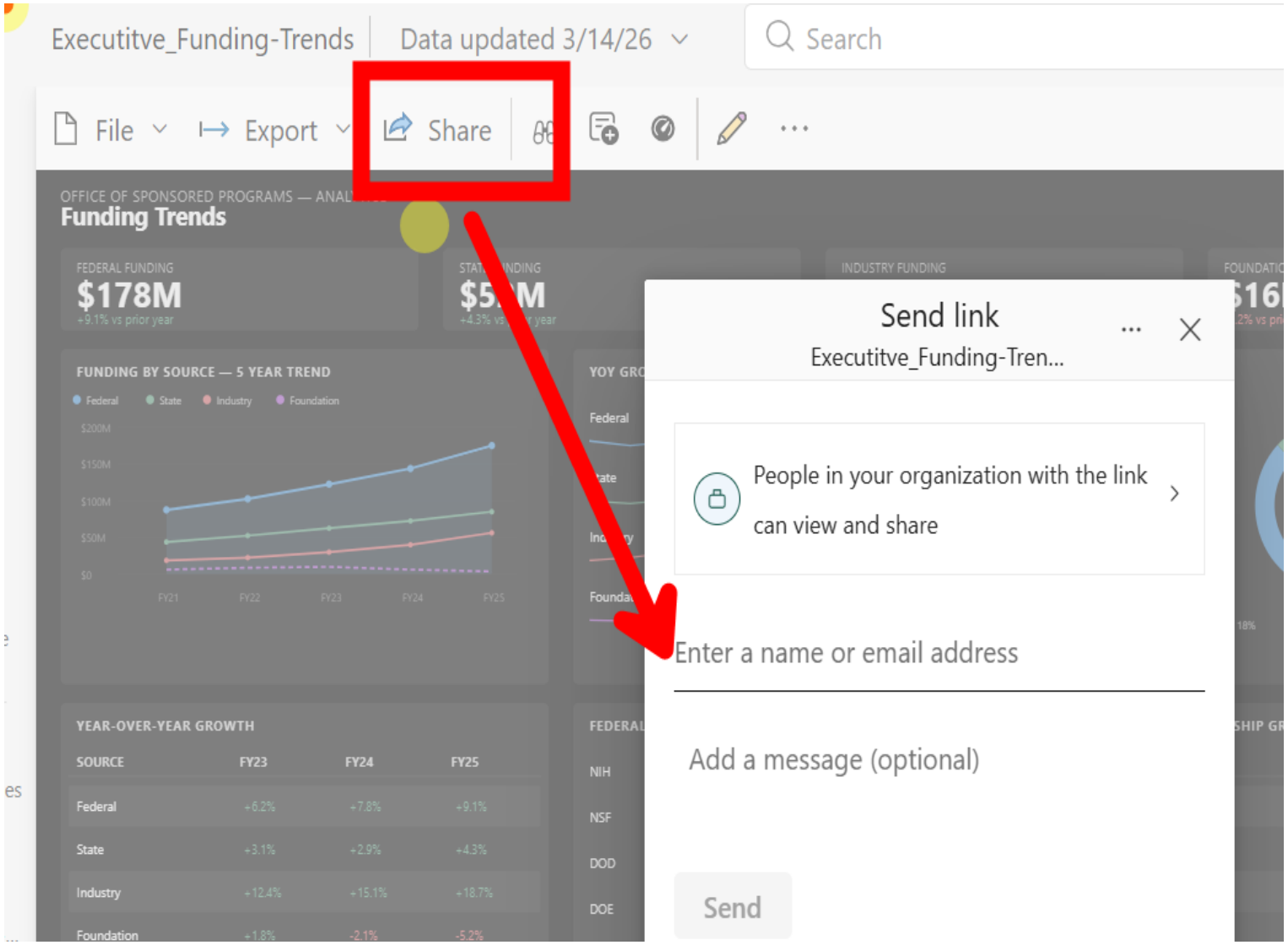
- Live demo next
- Power BI Desktop → click "Publish"
- Select your Workspace





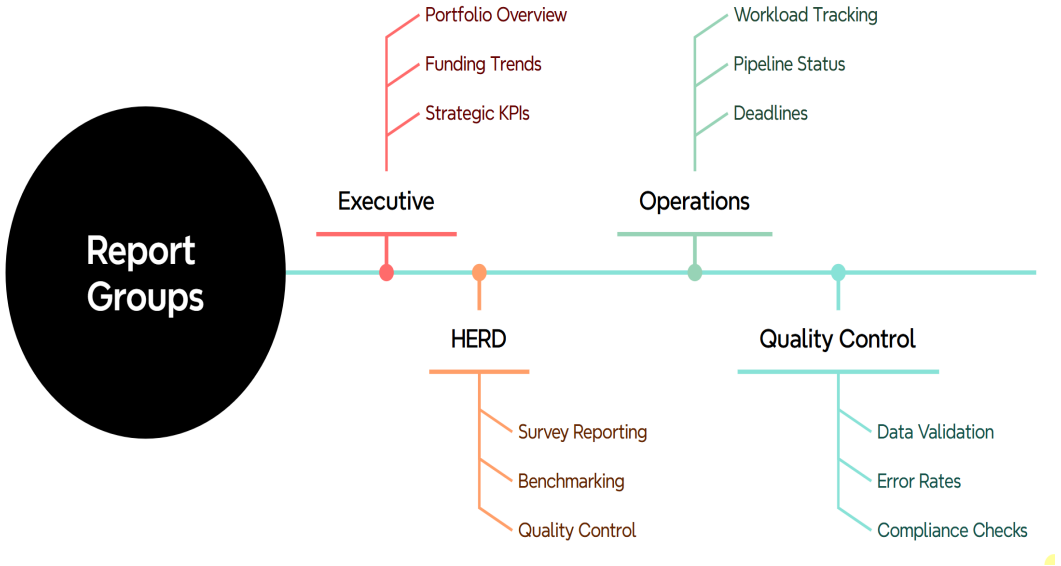
Direct Sharing via Workspace

- Individual user and report sharing
- Not scalable
- Hard to manage for large groups
- Let's look at an example



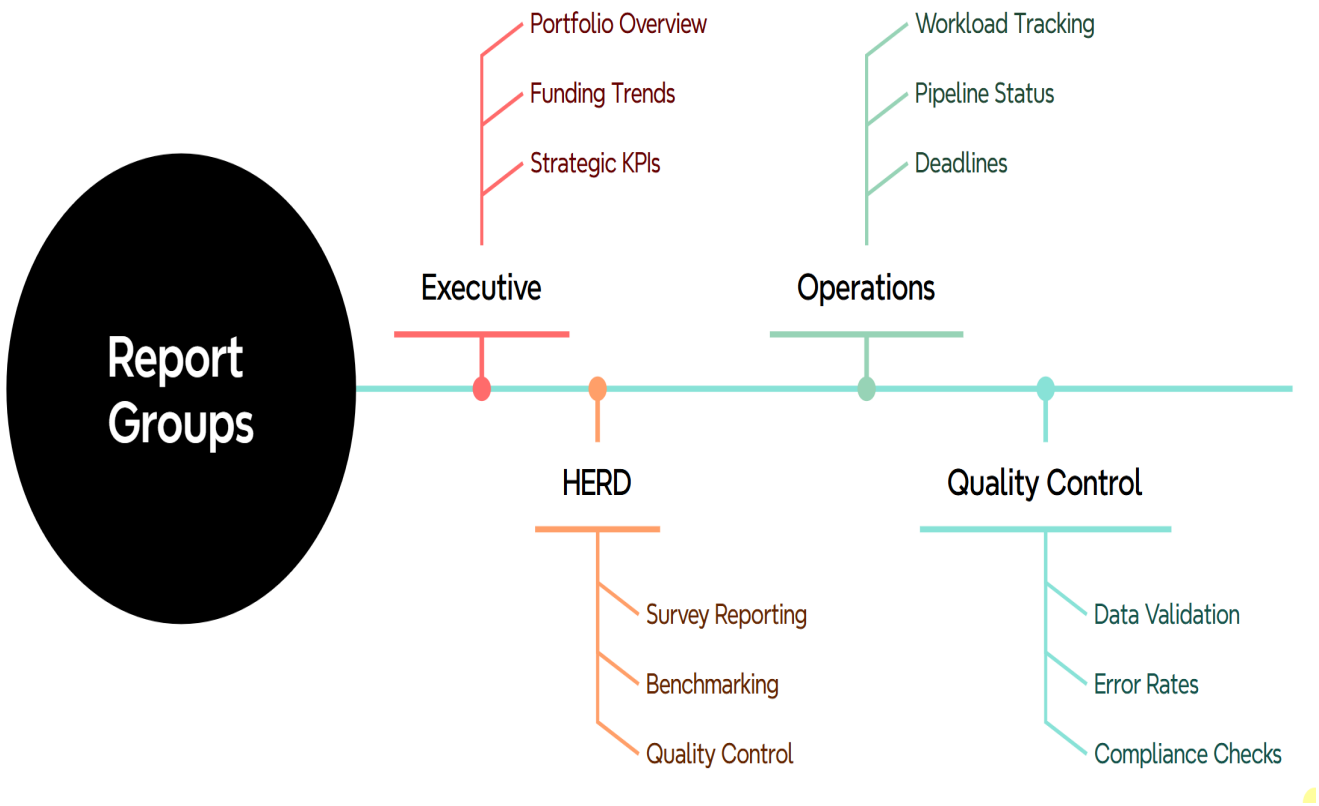
Example Problem

- Executive Reports: 3 reports, 10 users
- HERD Reports: 3 reports, 20 users
- Operations Reports: 3 reports, 10 users
- Quality Control Reports: 3 reports, 10 users
- 150 individual shares to manage!!



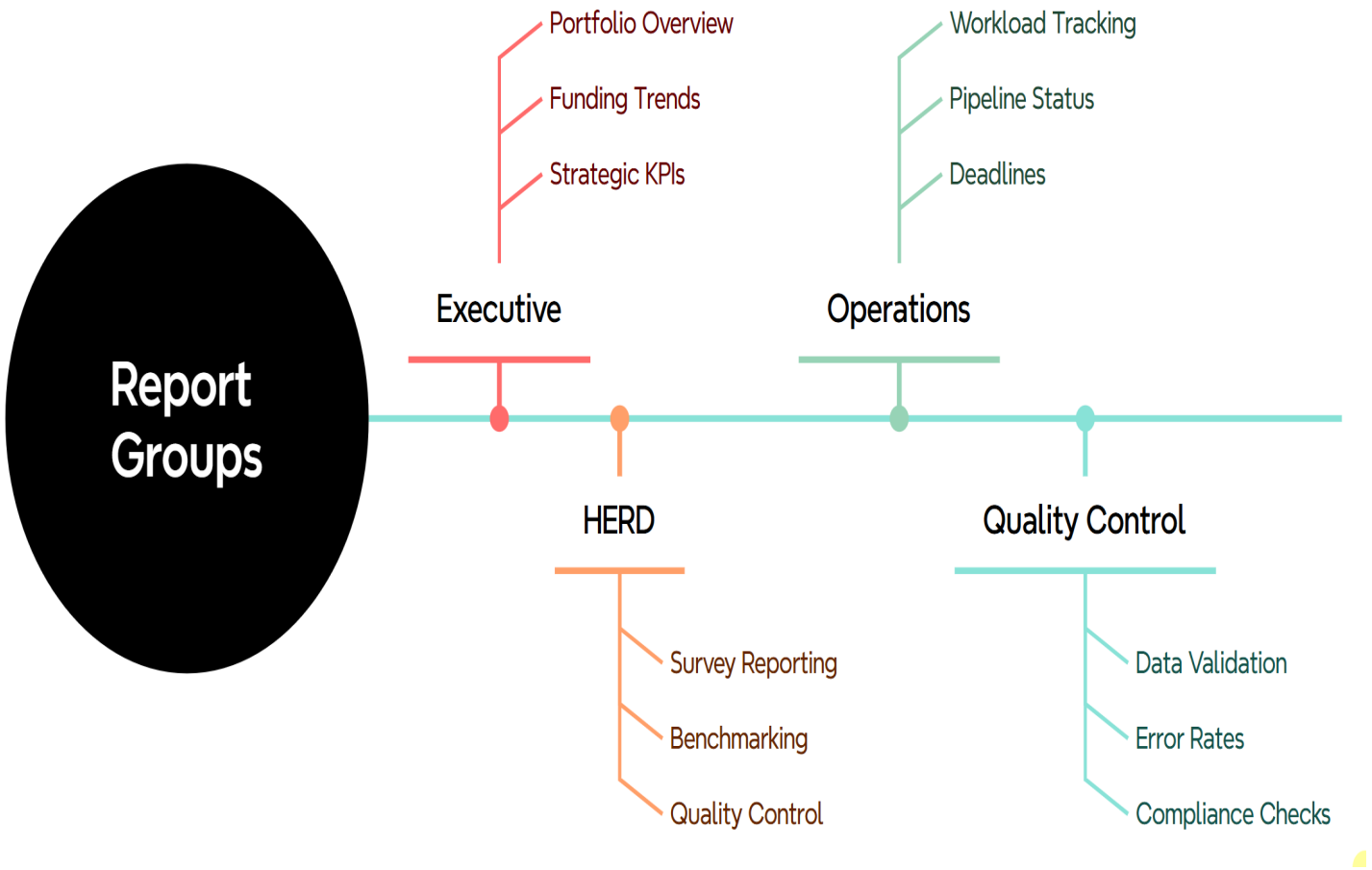
From Workspace to APP

- Defining who sees what within an App
- Audience groups
- Different pages to different users
- Managing access without duplicating reports



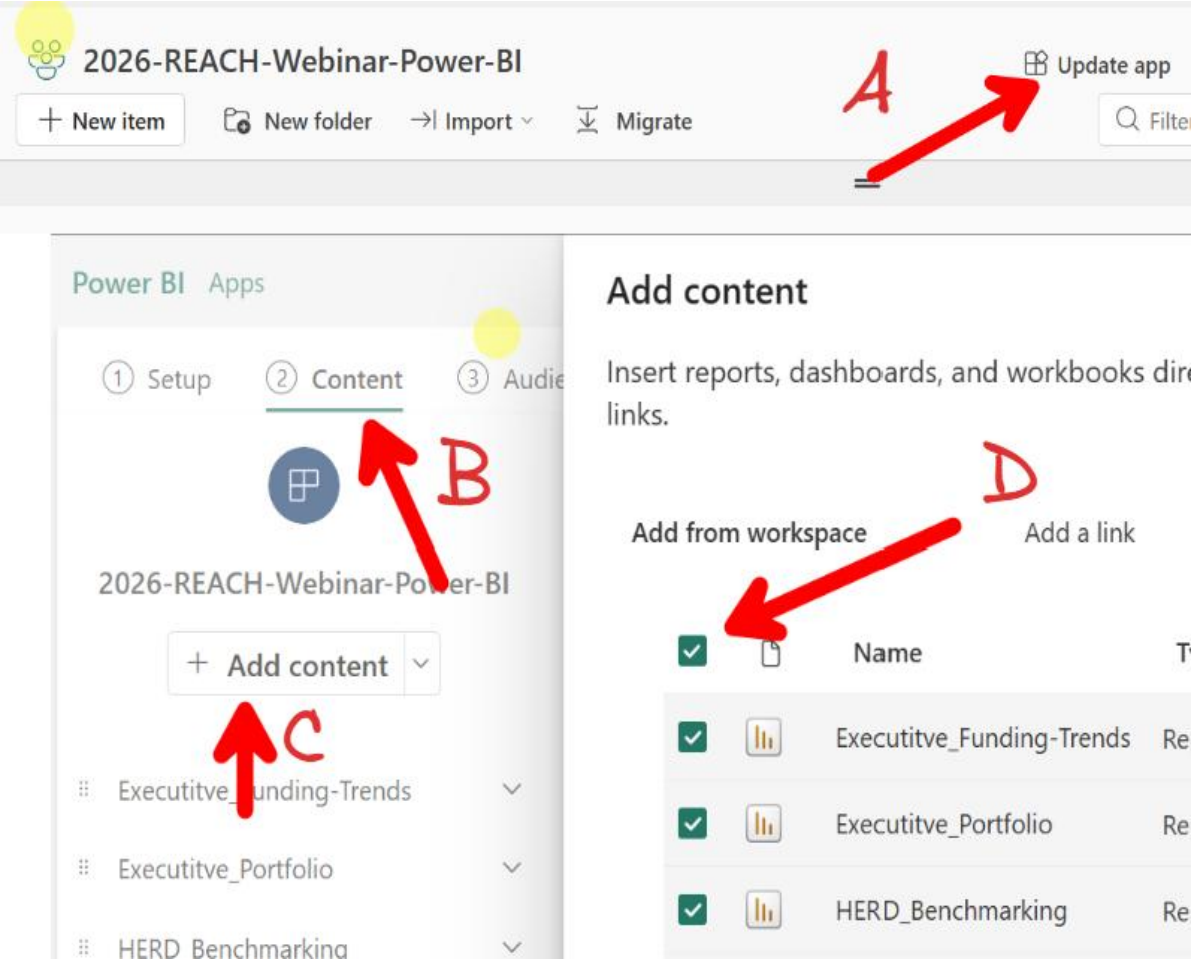
Report Groups

- Executive
- HERD
- Operations
- Quality Control



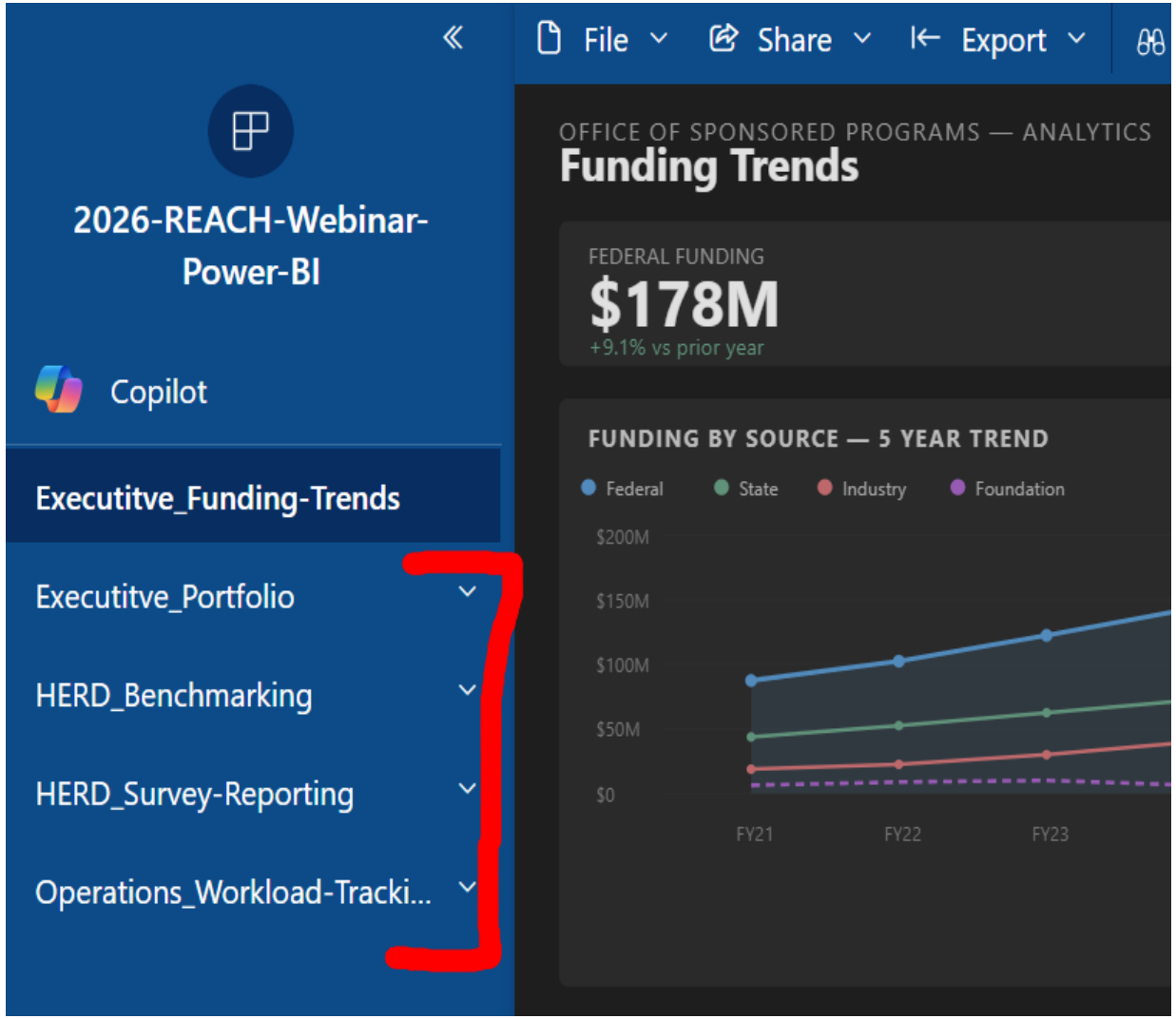
Deploying to an APP

- A. Click Create/Update APP
- B. Click "Content"
- C. Click "Add Content"
- D. Select reports

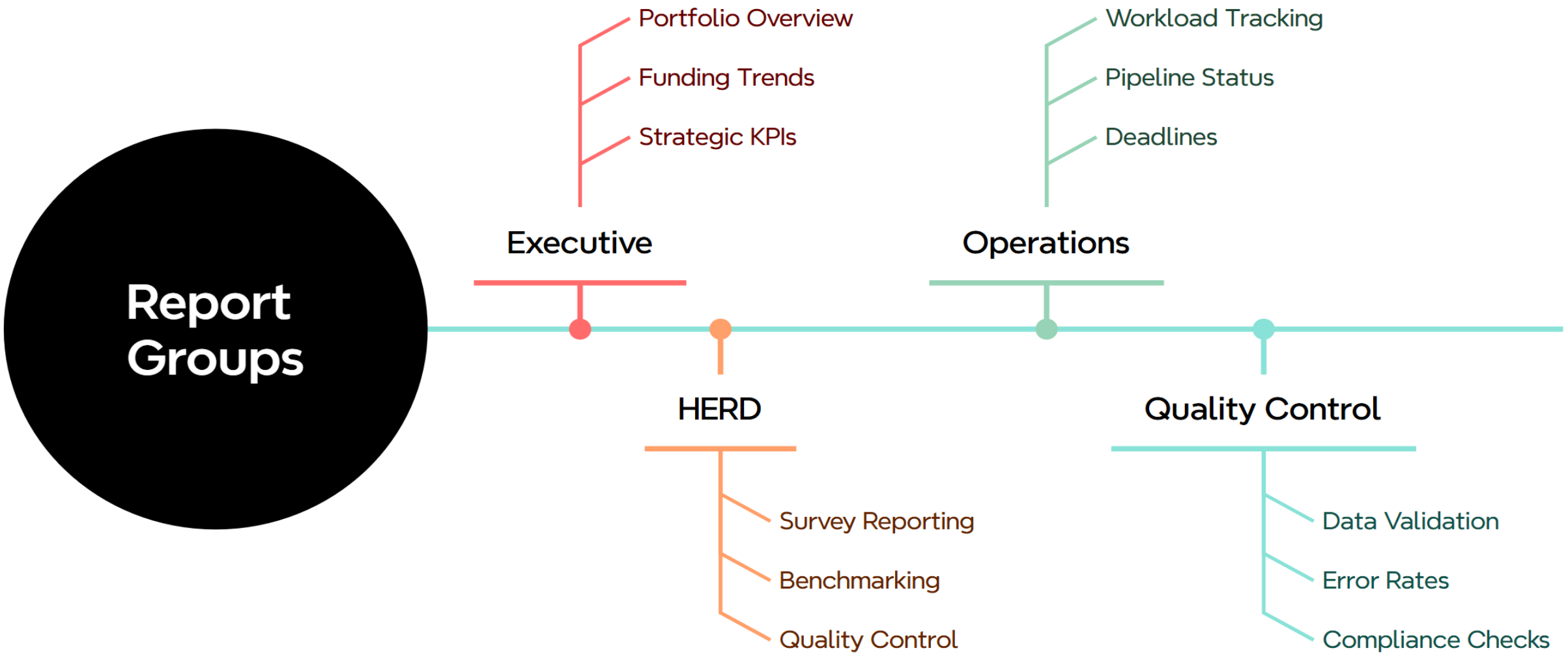


Split App into Reporting Groups

- App shows all reports
- This is not what we want

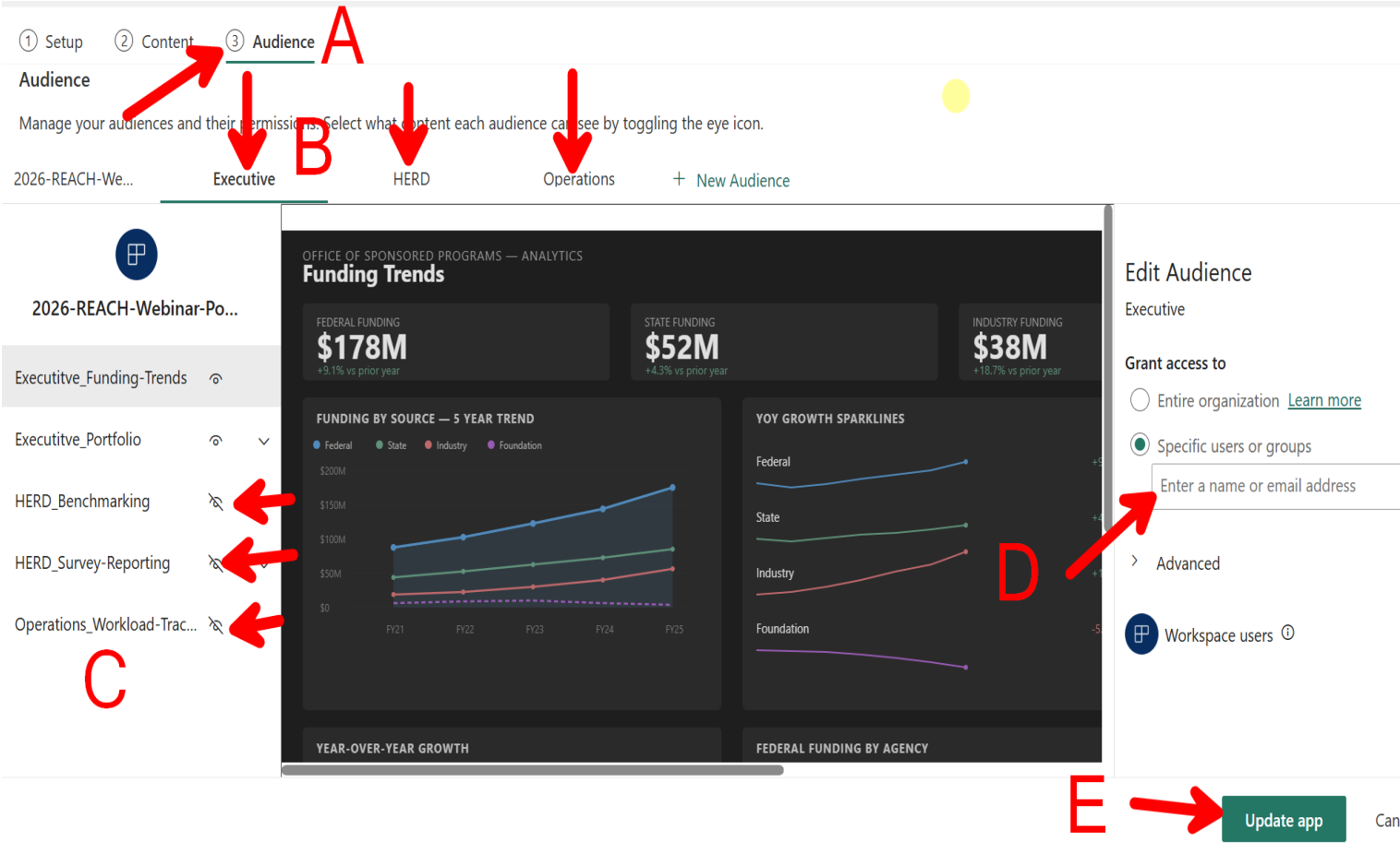


Reporting Groups

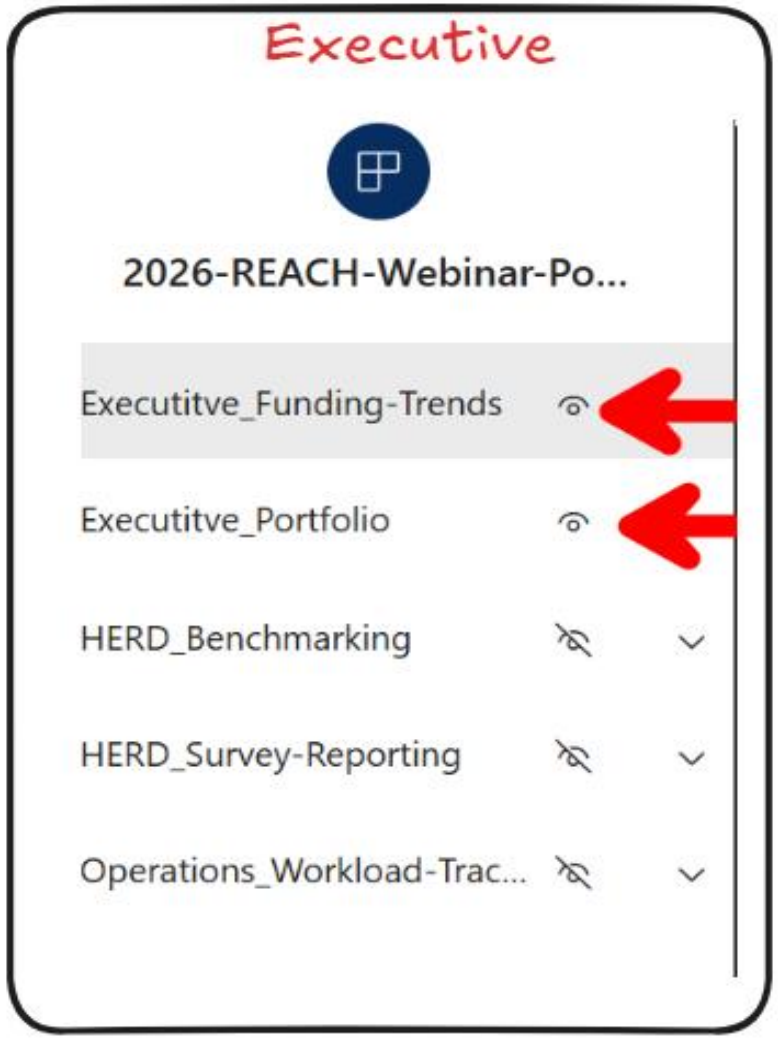


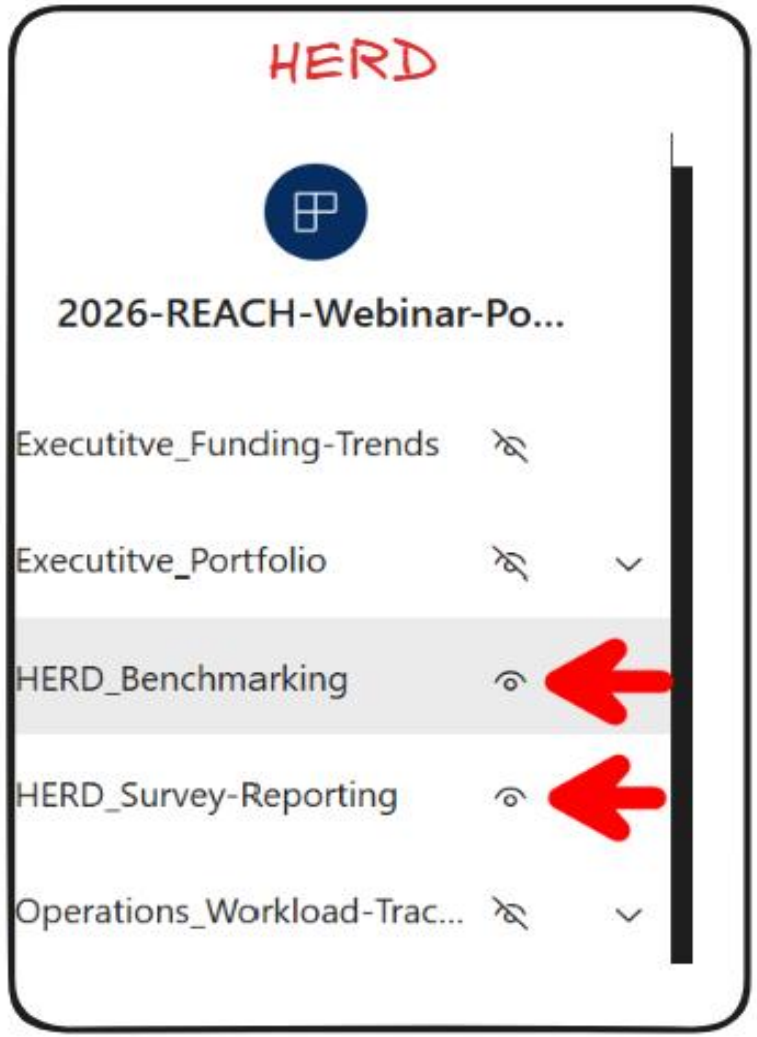
App Groups

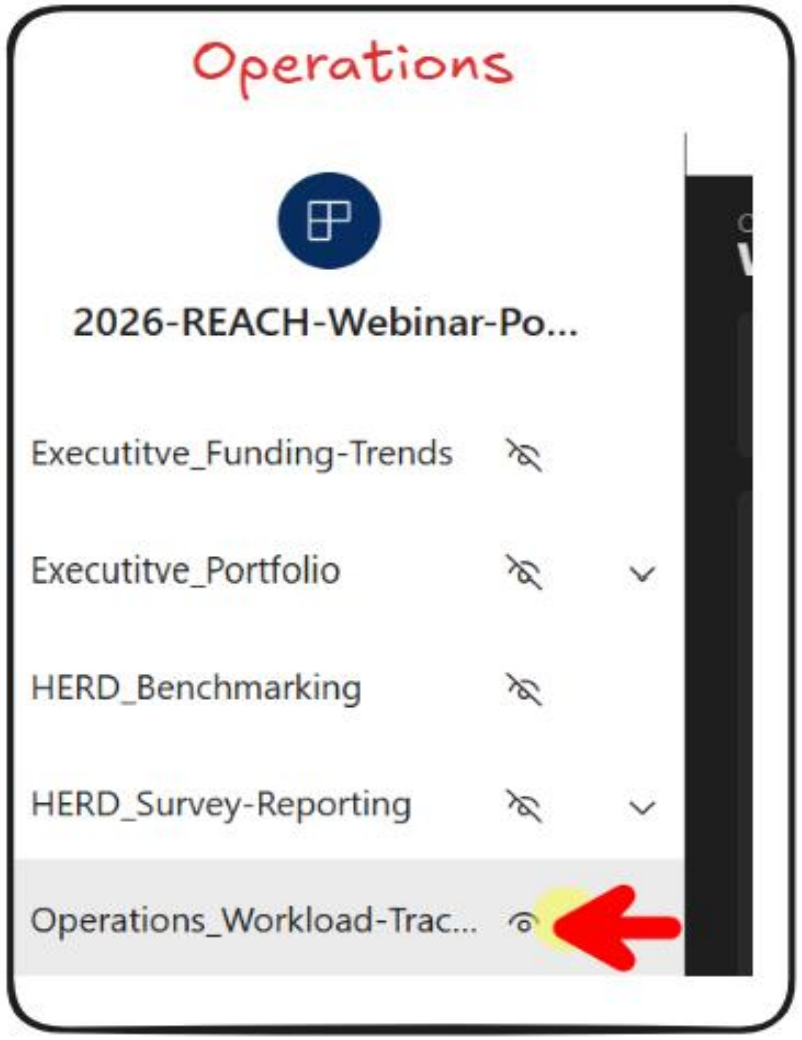
- Go back to workspace → click "Update App"
- A. Click "Audience"
- B. Create your groups
- C. Select reports for that group
- D. Add group members
- Individual users or Azure security groups



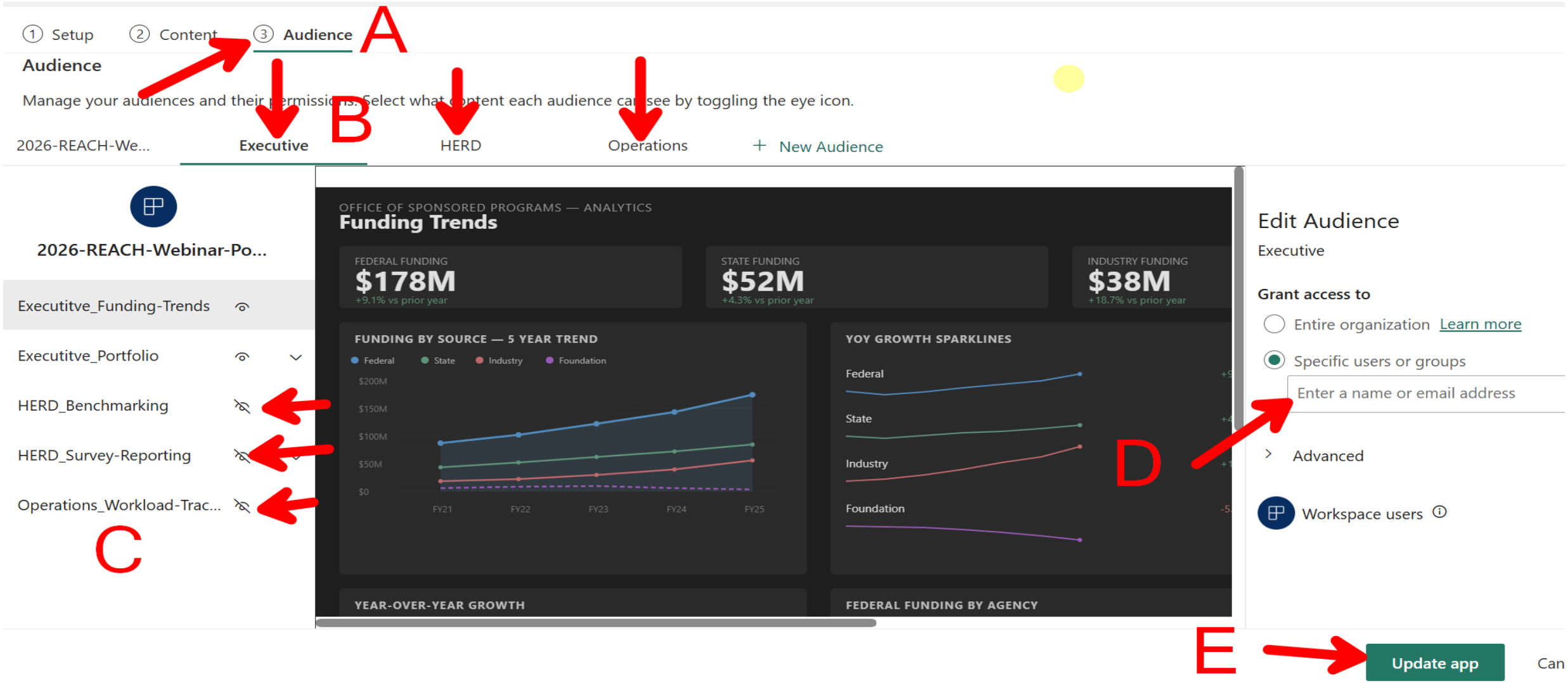
App Audience



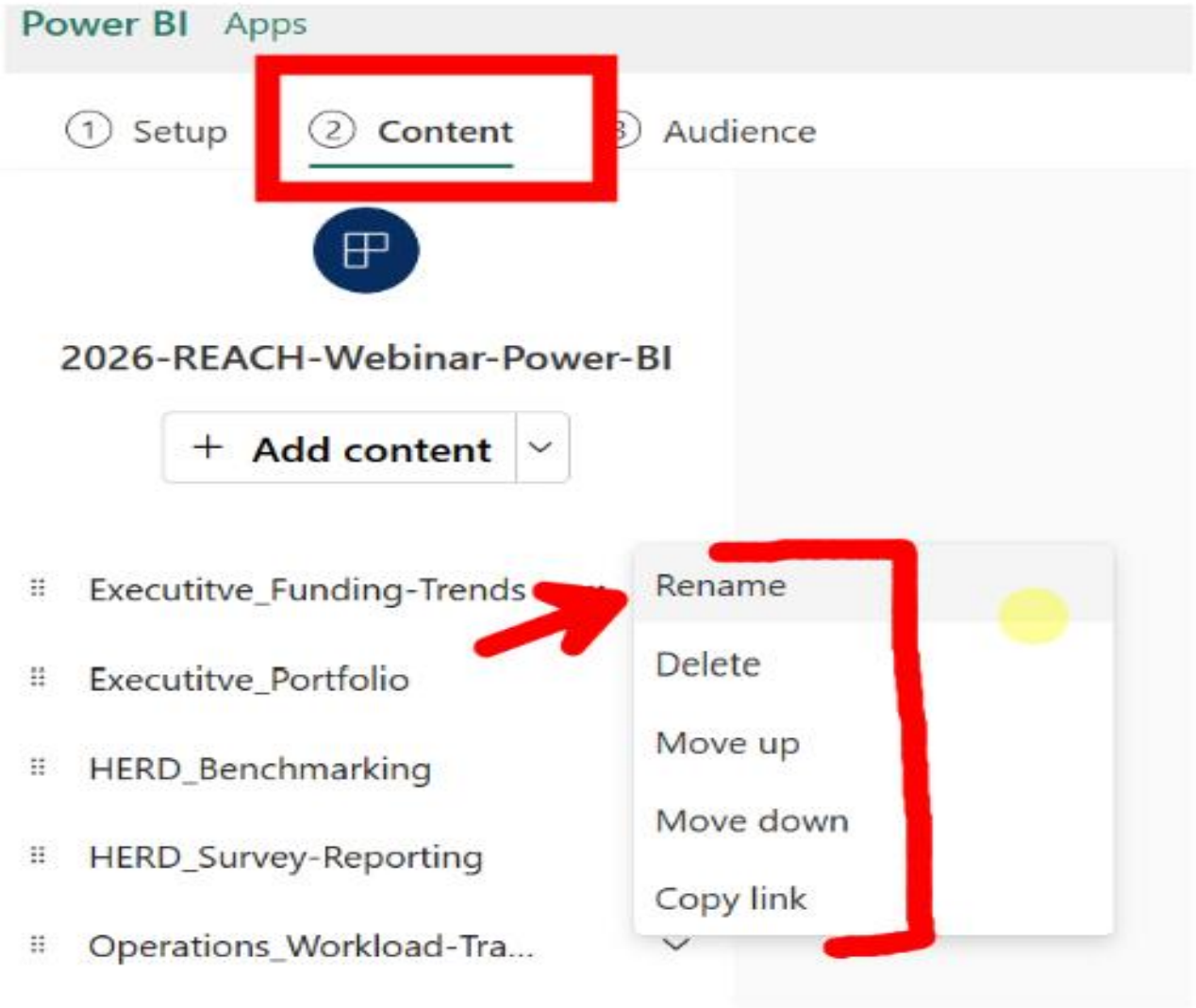


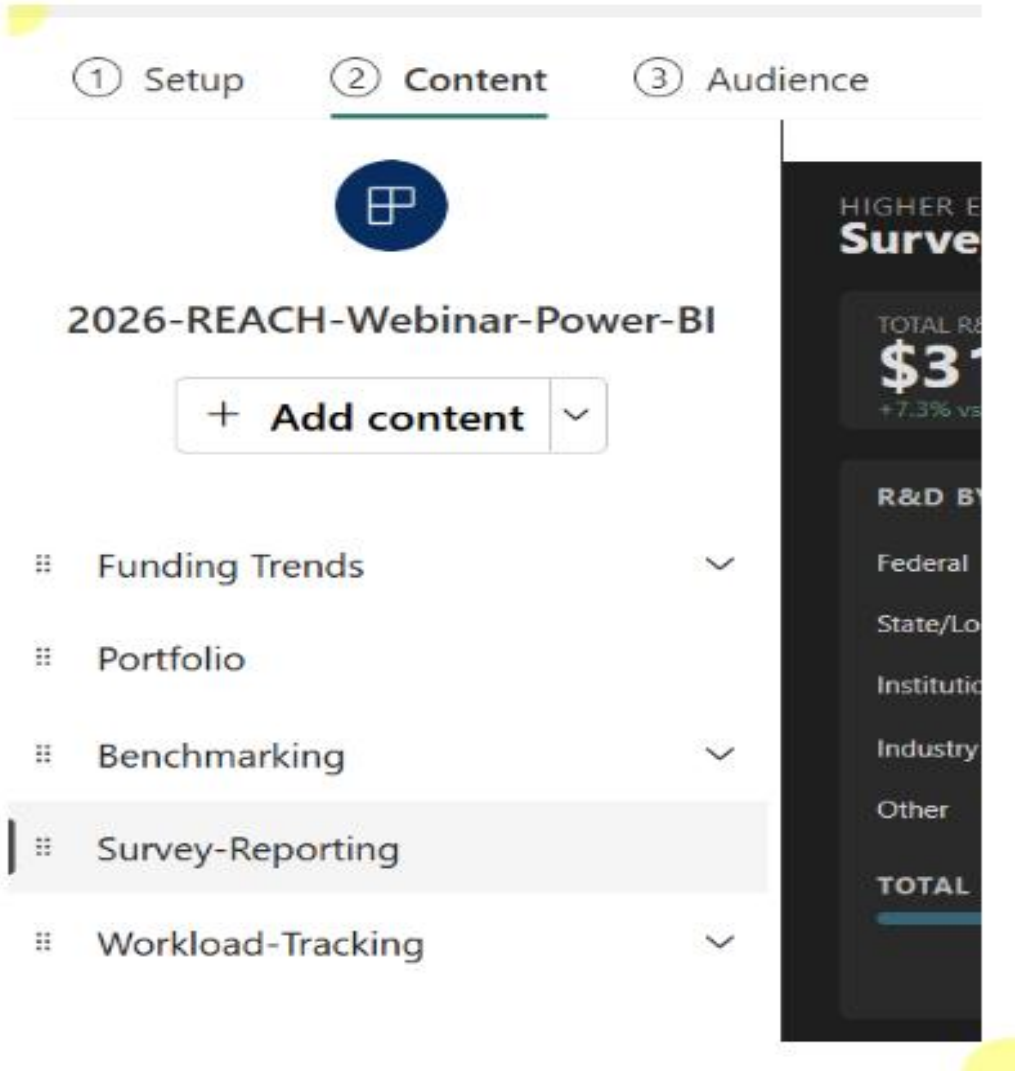


Live Demo

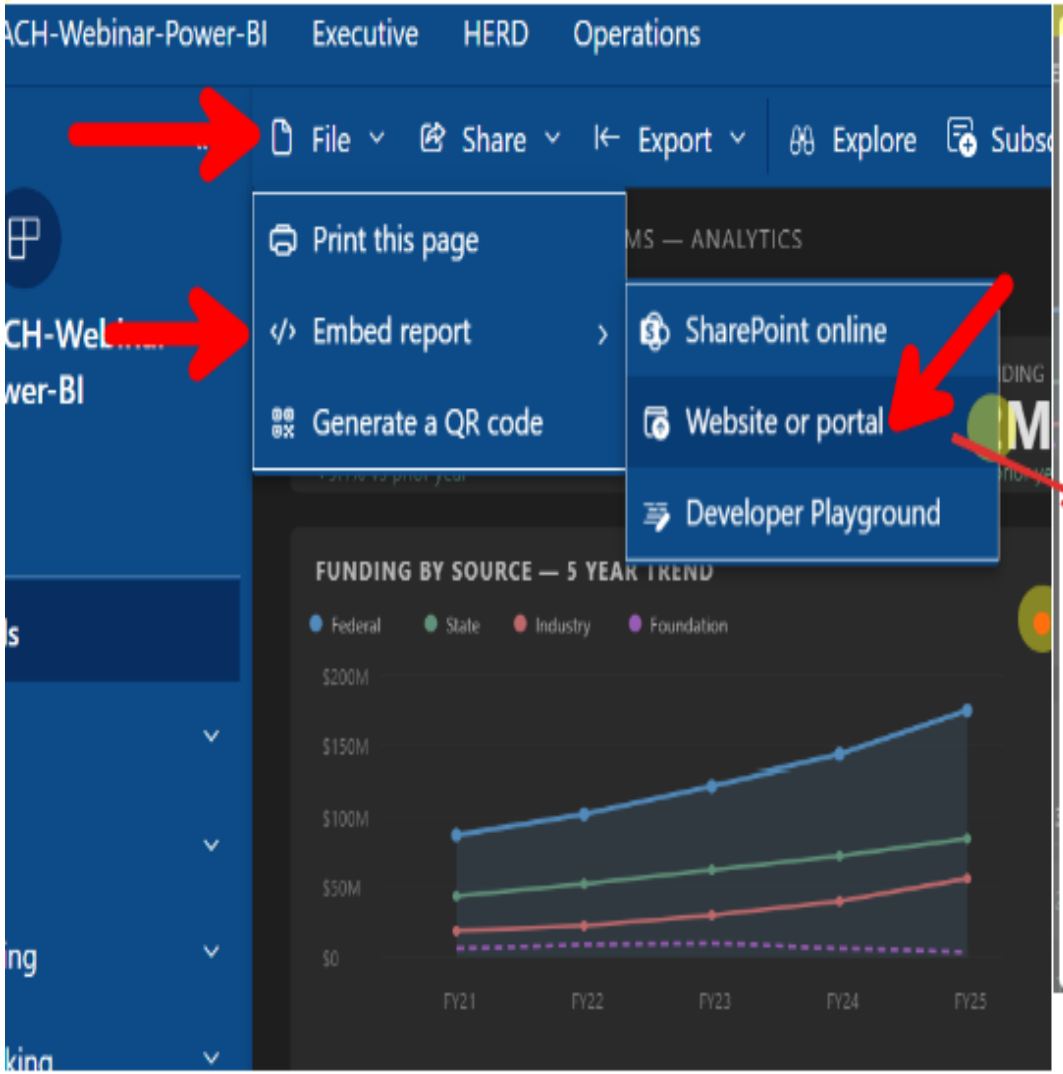

1 Setup
2 Content
3 Audience
A
Audience
Manage your audiences and their permissions. Select what content each audience can see by toggling the eye icon.
2026-REACH-We...
Executive
B
HERD
Operations
+ New Audience
2026-REACH-Webinar-Po...
Executitve_Funding-Trends
Executitve_Portfolio
HERD_Benchmarking
HERD_Survey-Reporting
Operations_Workload-Trac...
C
OFFICE OF SPONSORED PROGRAMS — ANALYTICS
Funding Trends
FEDERAL FUNDING
$178M
+9.1% vs prior year
STATE FUNDING
$52M
+4.3% vs prior year
INDUSTRY FUNDING
$38M
+18.7% vs prior year
FUNDING BY SOURCE — 5 YEAR TREND
Federal
State
Industry
Foundation
$200M
$150M
$100M
$50M
$0
FY21
FY22
FY23
FY24
FY25
YOY GROWTH SPARKLINES
Federal
State
Industry
Foundation
YEAR-OVER-YEAR GROWTH
FEDERAL FUNDING BY AGENCY
Edit Audience
Executive
Grant access to
Entire organization
Learn more
Specific users or groups
Enter a name or email address
D
Advanced
Workspace users
E
Update app
Can

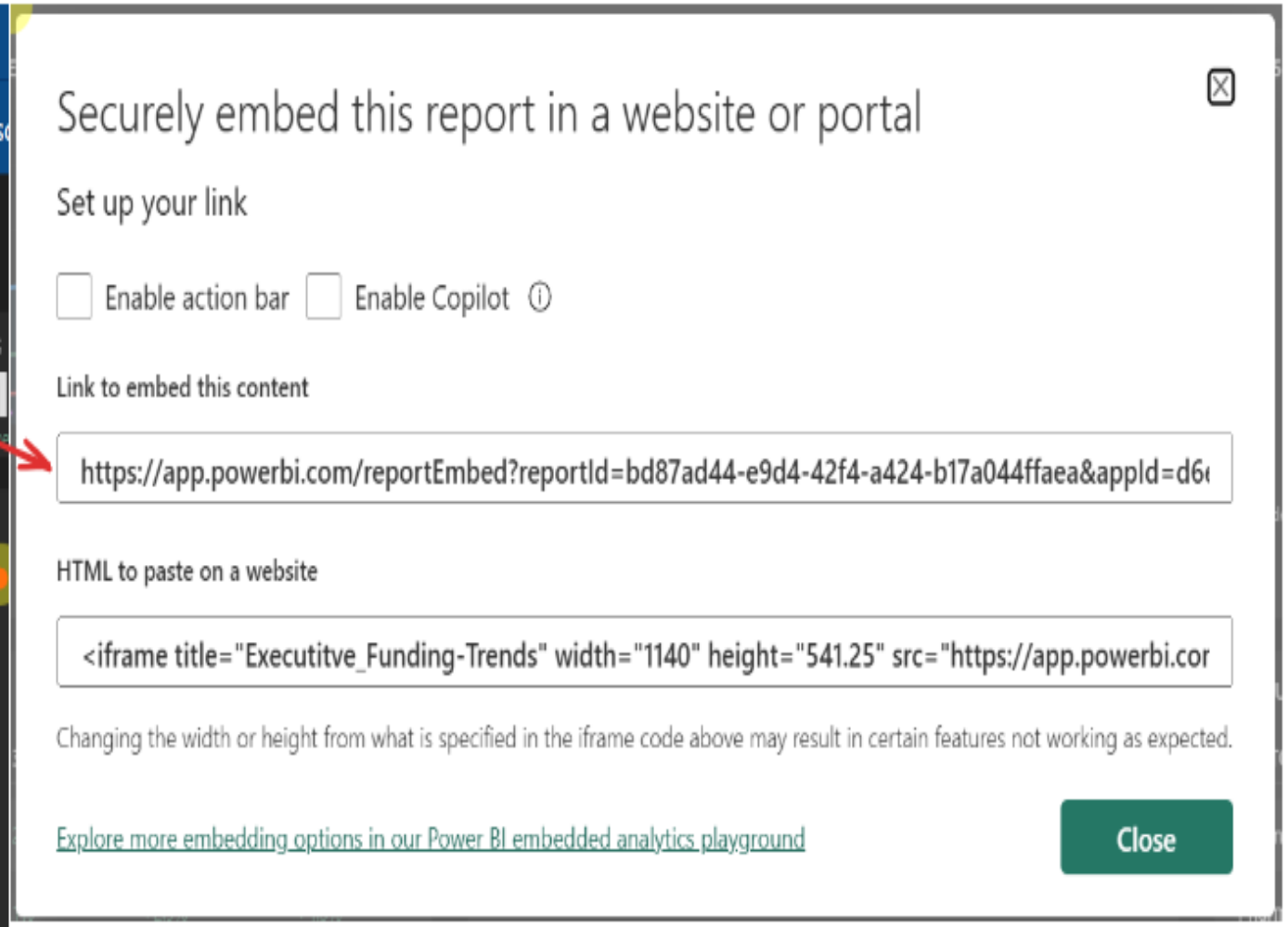
Renaming Report in the App





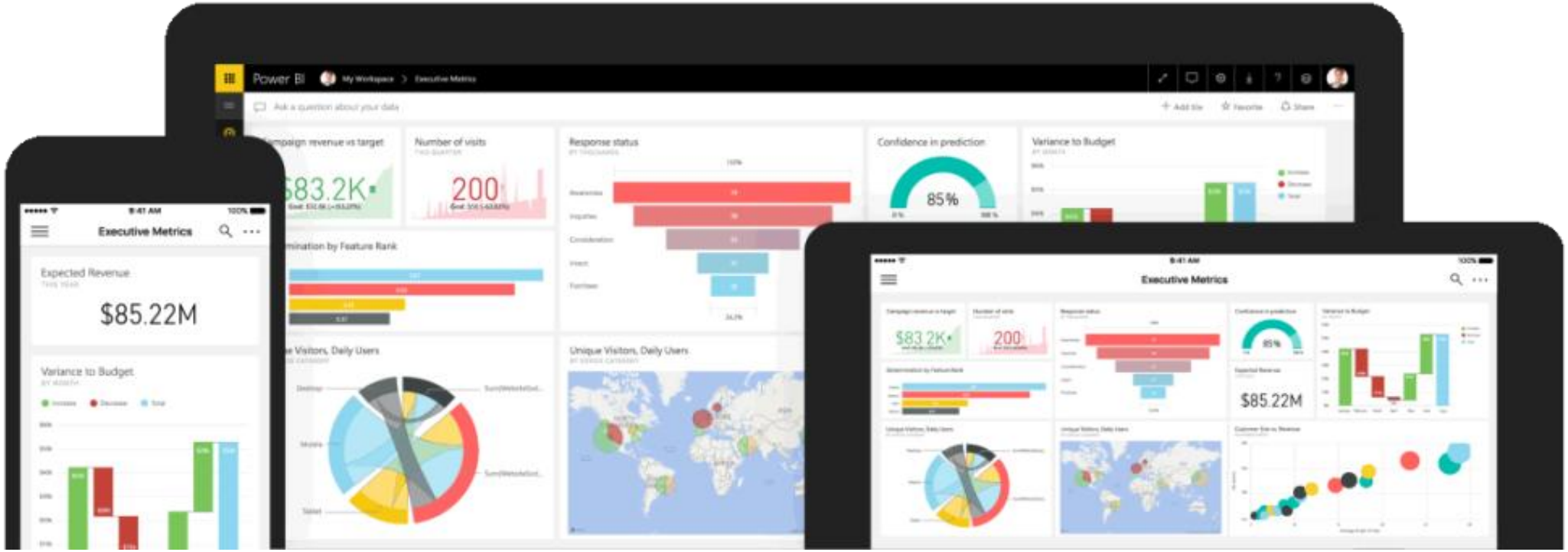
Embedding Reports in SharePoint





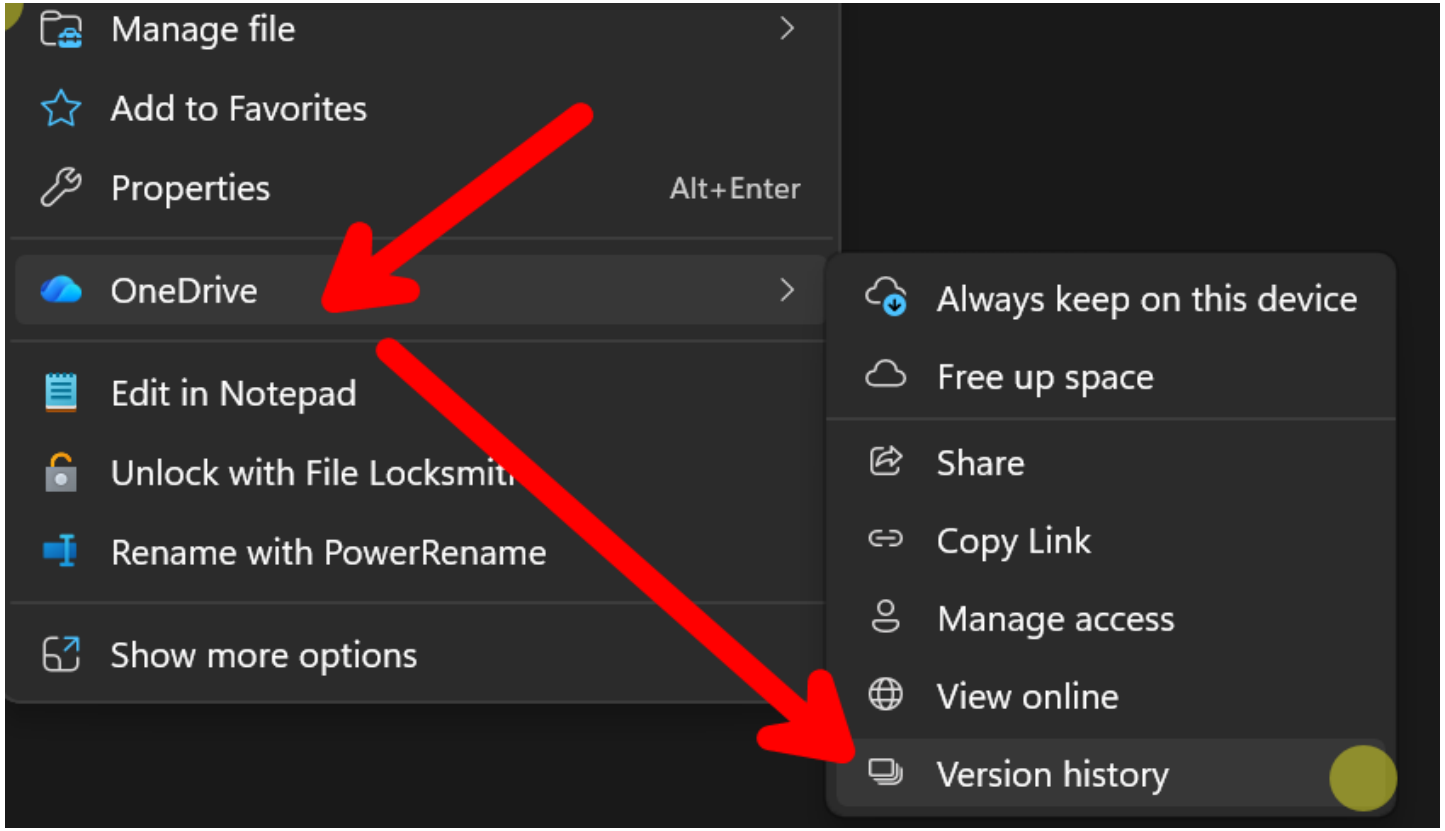
Power BI Mobile

- Customize views in the Power BI Desktop



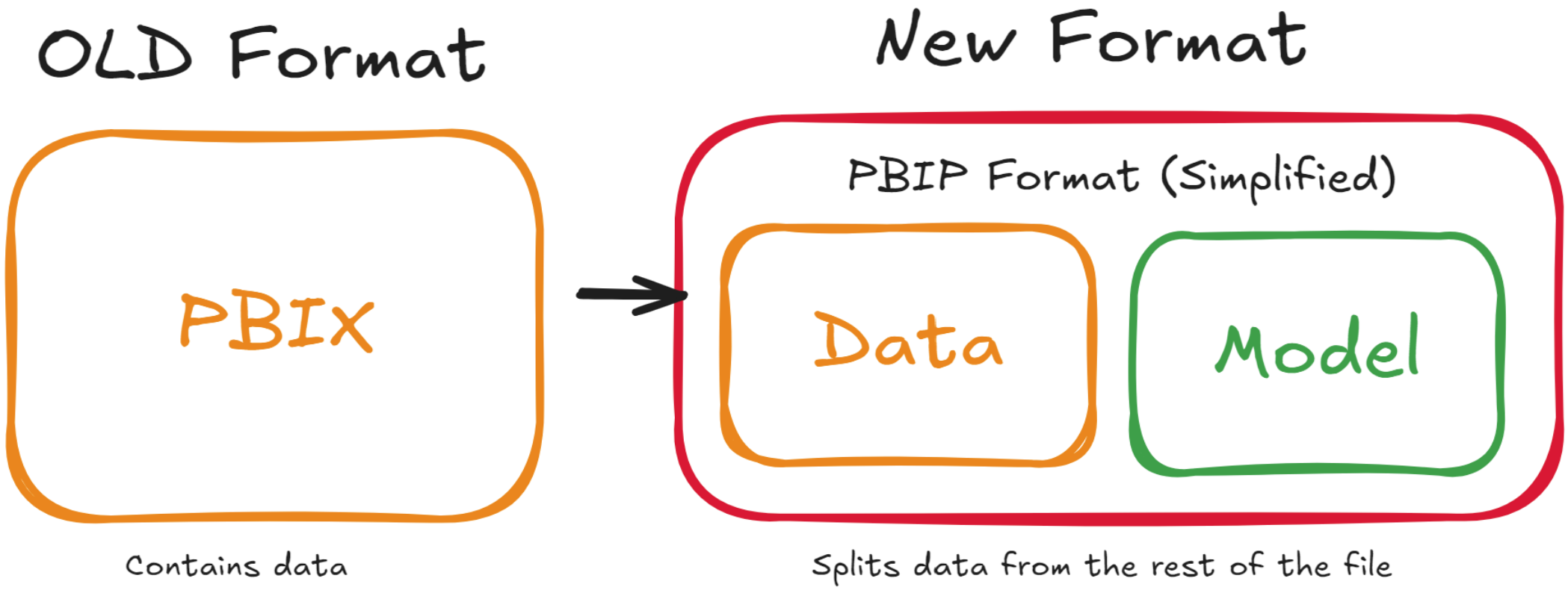
Version Control (Simple)

- By default all your work is already version controlled



Version Control Power BI (Advanced)

- Power BI PBIX vs PBIP format

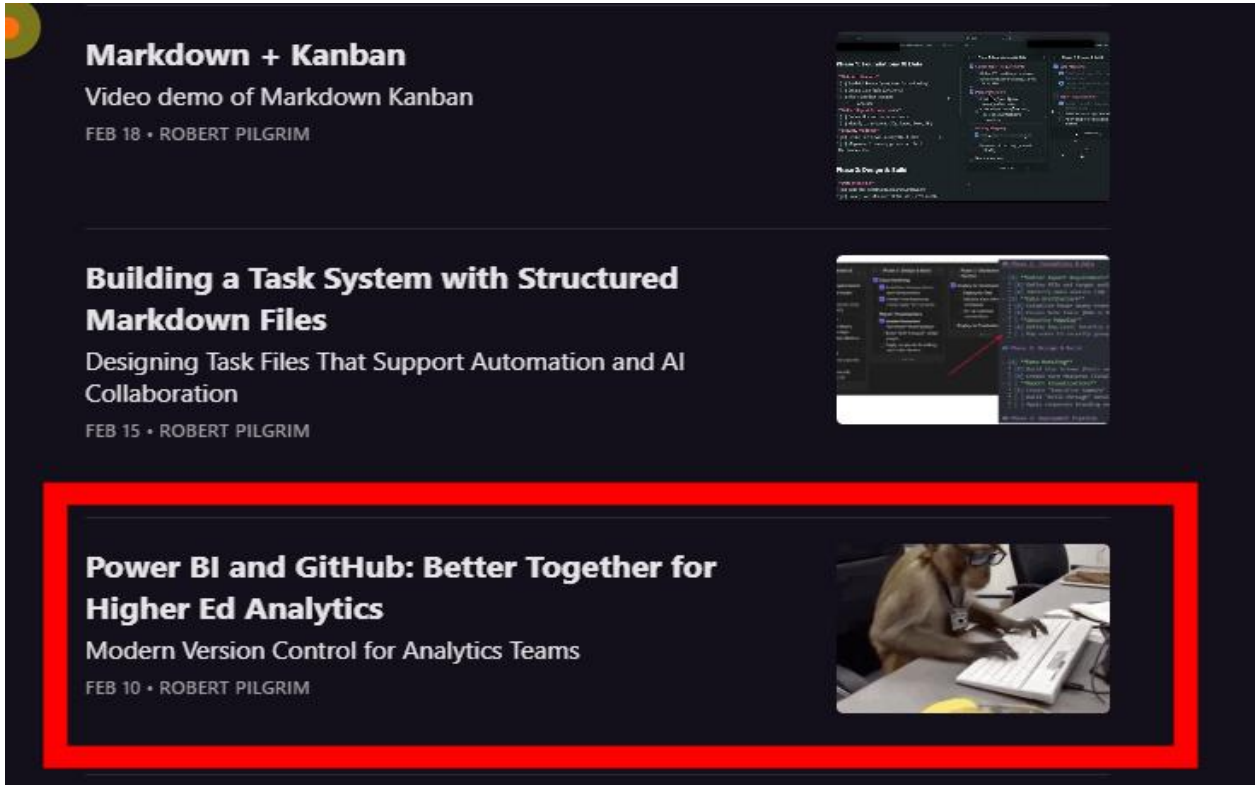


Version Control Power BI (Advanced)

- PBIP file format
- Data is not stored

Project_Folder			
1) MyProject.pbip	2) .gitignore	3) MyProject.Report (THE VISUAL LAYER)	4) MyProject.SemanticModel (THE DATA LAYER)
Main project entry file	Auto-generated for Git users	definition.pbir — Connection to the semantic model report.json — Legacy: All visuals in one file definition/ — Modern PBIR: Split visual files definition/ — pages/ — Individual folders per report page definition/ — version.json StaticResources/ — Images, icons, and themes	definition.pbir — Dataset configuration model.bim — Legacy: Entire model in one JSON definition/ — TMDL: Human-readable text files definition/ — tables/ — One .tmdl file per table definition/ — expressions.tmdl — Power Query / M code definition/ — culture/ — Translations and metadata .pbi/ — Local cache (Data) and user settings

Learn more from my Articles

- Power BI and GitHub: Better Together for Higher Ed Analytics

- Managing Complex Analytics Workflows in GitHub Using a Live Microsoft Repository